

Kye Burchard

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EDUCATION

MIT | SB ELECTRICAL ENGINEERING AND COMPUTER SCIENCE

GPA: 5.0

2017 - 2021

EXPERIENCE

Falconer | FOUNDING ENGINEER

Mar. 2025 - Present

- Built agents that autonomously maintain and update company knowledge bases — document operations, failure detection, and self-healing across the full knowledge lifecycle
- Owned entire product end-to-end: agent systems, retrieval infrastructure, streaming chat UI, web application, cloud and on-prem deployments

Stripe | STAFF SOFTWARE ENGINEER (API SERVICES)

Aug. 2024 - Mar. 2025

- Architected v2 API platform spanning SDKs, CLI, and docs for Stripe's entire API surface, including a new versioning scheme and release cadence, while preserving interoperability with v1
- Maintained 99.9999% availability for API gateway serving all external requests
- Designed and led engineering onboarding sessions, including 'Big Diagram of Stripe,' training hundreds of engineers on system architecture, API design, and debugging production systems

Stripe | SENIOR SOFTWARE ENGINEER (API SERVICES)

Jan. 2023 - Aug. 2024

- Defined company-wide API design patterns and guidelines — domain modeling, composability, and abstraction layering — used by Stripe's API Review process to evaluate every public API
- Designed and shipped key parts of GraphQL architecture (auth, batching, rate limiting, reliability)

Stripe | SOFTWARE ENGINEER (DASHBOARD PLATFORM)

Feb. 2021 - Dec. 2022

- Built automated performance tooling — detecting unused API fields, network waterfalls, and N+1 queries across hundreds of Dashboard pages
- Established reliability and security standards for Stripe's Dashboard, building automated error alerting and component ownership enforcement across team boundaries

HackMIT | LOGISTICS HEAD 2019, DEV & CORPORATE RELATIONS 2018

April 2018 - Sept. 2019

- Headed a 12-person team to plan and execute logistics for a 1,200-person, 3-day hackathon

Other

- **MIT Media Lab** Designed data pipeline to improve data quality & location detection for Rhythm Badges
- **web.lab** Coordinated logistics, staff, and delivered lectures for 200+ person web development course
- **Stripe Internship** Improved the performance of the highest-traffic page by 2-3x
- **Microsoft Garage** Designed & implemented a modular, scalable audit logging architecture as intern
- **Maseeh Hall** Led largest dorm at MIT in reviving culture, resulting in a 20% improvement in retention
- **SIPB** Keyholder & contributor at MIT's student computing group
- **Debate** Policy Debate state champion (Montana, 2017), MIT debate team (2018-19)

COURSEWORK

Performance Engineering, AI, Machine Learning, Computer Security, Operating Systems, Computer Systems Engineering

SKILLS

ADVANCED

TypeScript, Python, Go, Ruby, Java, AWS, Next.js, Linux, C, HTML/CSS, gRPC/Protobuf

KNOWLEDGEABLE

Terraform, Pulumi, C++, C#, PostgreSQL, GCP, Rust, SystemVerilog, $\LaTeX$